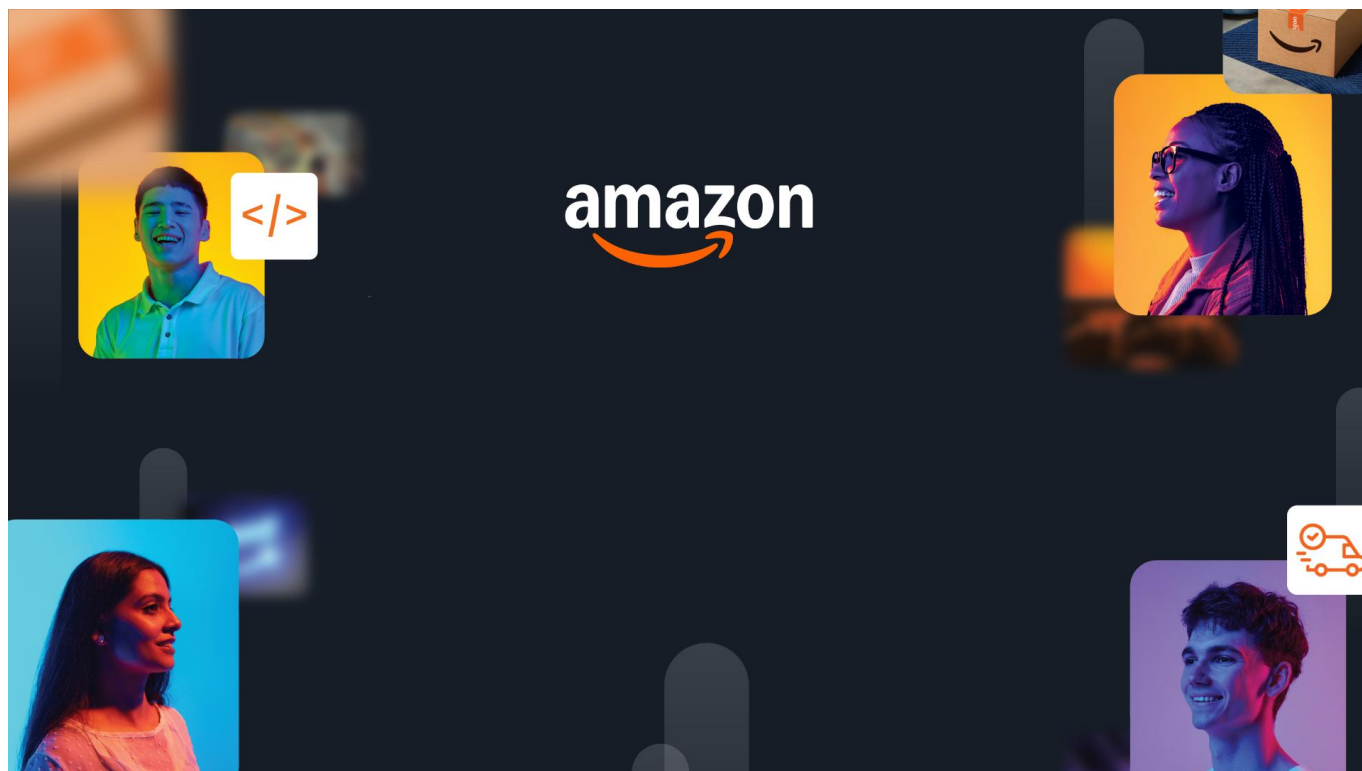




HackOn with Amazon

A Universe of Opportunity

48-Hour Hackathon | Solution Document

Team Name	Powerpuff Girls
Hackathon Theme	Amazon Now – Reimagining Urgent Shopping
Date	15 th June 2026

Team Members

Name	College / University	Role	Email
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1. Problem Statement & Relevance

The Problem

Modern e-commerce has solved product availability but not decision-making. Customers are expected to evaluate product quality, seller reliability, review authenticity, sustainability impact, compatibility with their needs, and purchase confidence within seconds. As a result, shoppers frequently experience **decision fatigue, purchase regret, unnecessary returns, and abandoned carts** despite having access to vast amounts of information.

Why It Matters

This challenge affects **hundreds of millions of online shoppers** globally and represents a significant cost for both consumers and marketplaces. Poor purchasing decisions lead to **higher return rates, wasted logistics resources, lower customer trust, and increased shopping friction**. In fast-paced shopping environments where customers arrive with immediate intent, every additional second spent validating a decision increases the likelihood of abandonment or dissatisfaction.

Theme Alignment

Amazon Lens reimagines urgent shopping **by transforming product discovery into decision certainty**. Instead of requiring customers to manually search, compare, verify, and evaluate products, Amazon Lens acts as *an intelligent decision layer* that helps users discover what they need, understand whether they can trust it, and purchase with confidence in the shortest possible time. The platform **reduces cognitive load while accelerating the journey from intent to purchase**.

What Makes This Novel

While existing quick-commerce platforms focus solely on logistics (speed of delivery), our approach focuses on **cognitive speed** (speed of decision-making). We are introducing a **multi-dimensional ecosystem** that turns individual shopping into **a predictive, social, and hyper-transparent experience**:

- **Radical Transparency:** Consolidating deep backend data (return rates, reorder velocity, seller tenure) into a single, instant trust score, replacing the need to read reviews.
- **Social Synchronicity:** Introducing real-time peer-to-peer validation and collaborative cart-building for urgent group needs.
- **Predictive Cohesion:** Leveraging NLP and internal behavioural preferences to curate instant, contextual setups rather than isolated products.

2. Customer & Solution

Target Customer

The "Right-Now" Shopper. This is an on-the-go professional, busy parent, or host in the middle of an active event *who has an immediate, high-intent need* (e.g., a host running out of party supplies, or a remote worker whose laptop charger just fried). They *don't have time* to browse or compare options; they *need* **highly trustworthy, accurate products delivered instantly**, with absolute certainty and zero cognitive friction.

How We Solve It

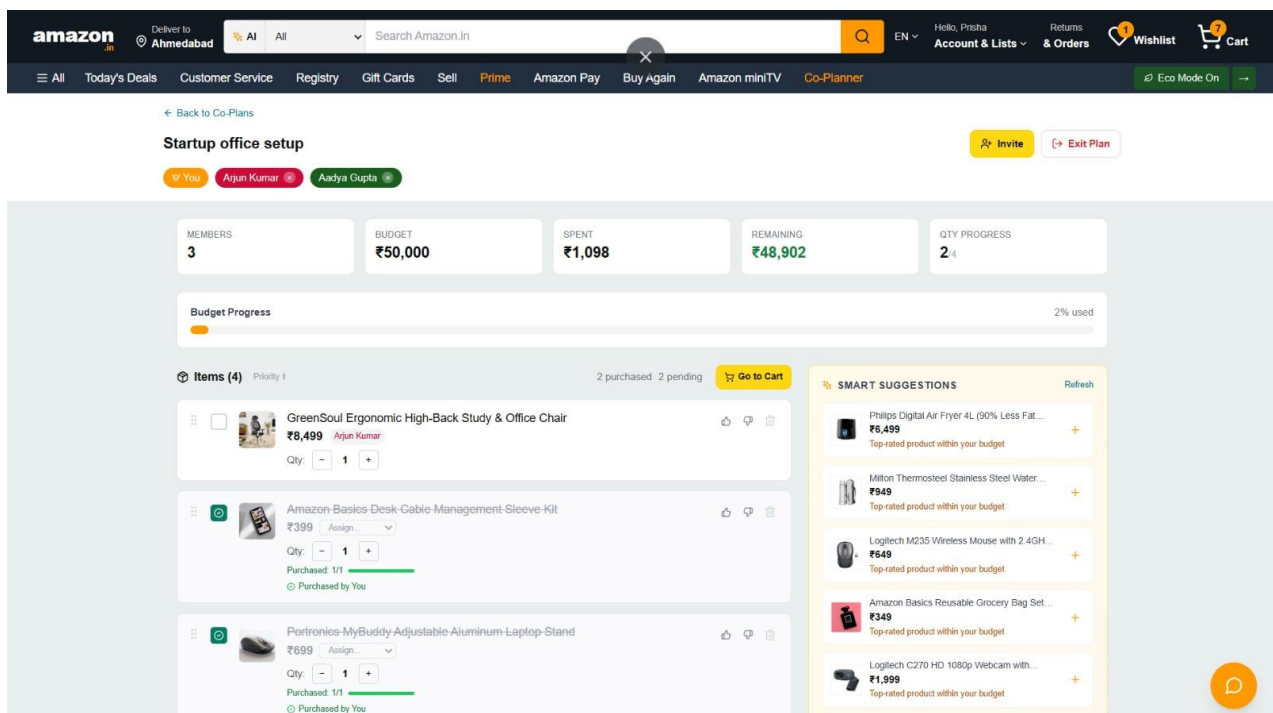
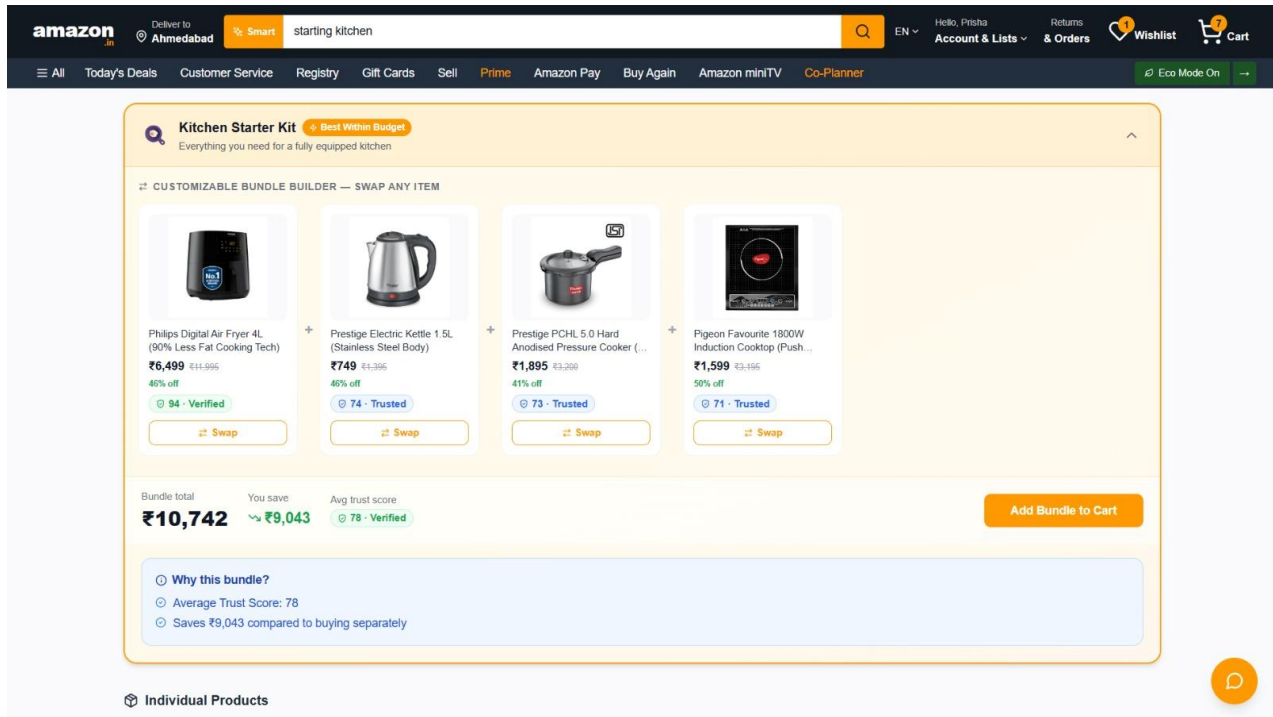
We strip the friction out of the shopping process by removing the need to read reviews, compare separate items, or coordinate orders over text threads. Our key features include:

- **TrustLens:** Eliminates review-skimming by consolidating Amazon's deepest backend data into a single, glanceable trust card. Right after clicking a product, users instantly see its true health: rating *across verified reviews, return rate, reorder rate, and the seller's track record*.
- **Smart Search & Predictive Bundles:** Replaces traditional keywords with natural language processing. If a user types "*need to study more*," the app bypasses individual item searches and suggests a *complete, single-click bundle*. Moreover, bundles are also displayed on the home page based on what a user may actually need and past purchase history.
- **Amazon Sense & EcoSafe:** Operates as an internal filter that automatically matches items to the user's unspoken preferences (like a history of buying organic or specific brands) and applies an EcoSafe sustainability score, eliminating the need to manually toggle filters. Additionally, Amazon Sense tracks historical purchase frequencies and consumption cycles to predict exactly when an item is running low. It proactively alerts the user via high-priority notifications before the deficit occurs, enabling a single-tap, friction-free reorder right when the need becomes urgent.
- **Witness & CoPlanner:** Solves group coordination and final purchase hesitation. CoPlanner lets multiple people add items to a single shared cart in real time (perfect for roommates or event hosts). If a buyer is still unsure, Witness opens a live, real-time chat with a verified past buyer who is online right now to get an immediate answer to a quick question.

User Workflow



Working Prototype



amazon **Deliver to Ahmedabad** AI All Search Amazon in EN Hello, Priya Account & Lists Returns & Orders Wishlist Cart

All Today's Deals Customer Service Registry Gift Cards Sell Prime Amazon Pay Buy Again Amazon miniTV Co-Planner Eco Mode On

Home > Electronics > Home Audio

JBL Cinema SB271 2.1 Channel Soundbar with Wireless Subwoofer

★★★★☆ 53 | 53 ratings | [Search this page](#)

TrustLens™
Predicted trust score: 72/100 **TRUSTED**

WHY BUYERS TRUST THIS SELLER

- 4.32 stars - 53 verified reviews**
Based on verified purchases in our customer database
- Only 8% of customers returned this**
Most customers keep this item
- 35% of buyers reordered this**
Customers keep coming back
- 98% on-time delivery**
Orders consistently arrive by the promised date
- 8-year seller - 427k+ orders fulfilled**
Established seller with a long track record on Amazon

Score = avg rating (35 pts) + low return rate (35 pts) + reorder rate (30 pts). Based on 50 verified buyers.

Was this seller trustworthy? 81 trust 28 flagged **Trust** **Flag**

☒ Simulate a return to affect the trust signal **Return this item**

89% Match
This product is a good fit based on your preferences.
✓ Similar to products you frequently purchase
✓ Matches categories you engage with often **AMAZON SENSE**

Share Wishlist

₹8,999
prime **FREE Delivery**
Get it by Tomorrow, 6 PM
Deliver to Ahmedabad
In Stock
Qty: 1
Add to Cart
Buy Now
Add to Co-Plan

Sold by **JBL Official Store**
Seller rating 4.7 ★
Seller since 2018
Fulfilled by Fulfilled by Amazon

☐ Amazon.in Return Policy
☒ Free delivery on orders above ₹499
☐ 10 days replacement guarantee

TrustLens™
TRUSTED **72**

amazon **Deliver to India** AI All Search Amazon in EN Hello, Vandita Account & Lists Returns & Orders Wishlist Cart

All Today's Deals Customer Service Registry Gift Cards Sell Prime Amazon Pay Buy Again Amazon miniTV Co-Planner Eco Mode On

Home > Account > Sustainability

Your Sustainability Dashboard

Updated 06:04 PM **Eco Mode: On**

Your Sustainability Score
66
Moderate
Based on 50 purchases
Industry avg: 54

Eco-Friendly Purchases
5
Score 9/70

Sustainable Packaging
20%
of your purchases

Categories Shopped
9
distinct categories

Eco-Certified Items
11
certified products

SUSTAINABILITY PERSONALITY
Balanced Buyer
Your average score is 66. 20% of your purchases use sustainable packaging. Small switches could push your score past 70.

Sustainability Evolution
Average eco score per month from your orders
Current: 66
Jun 42

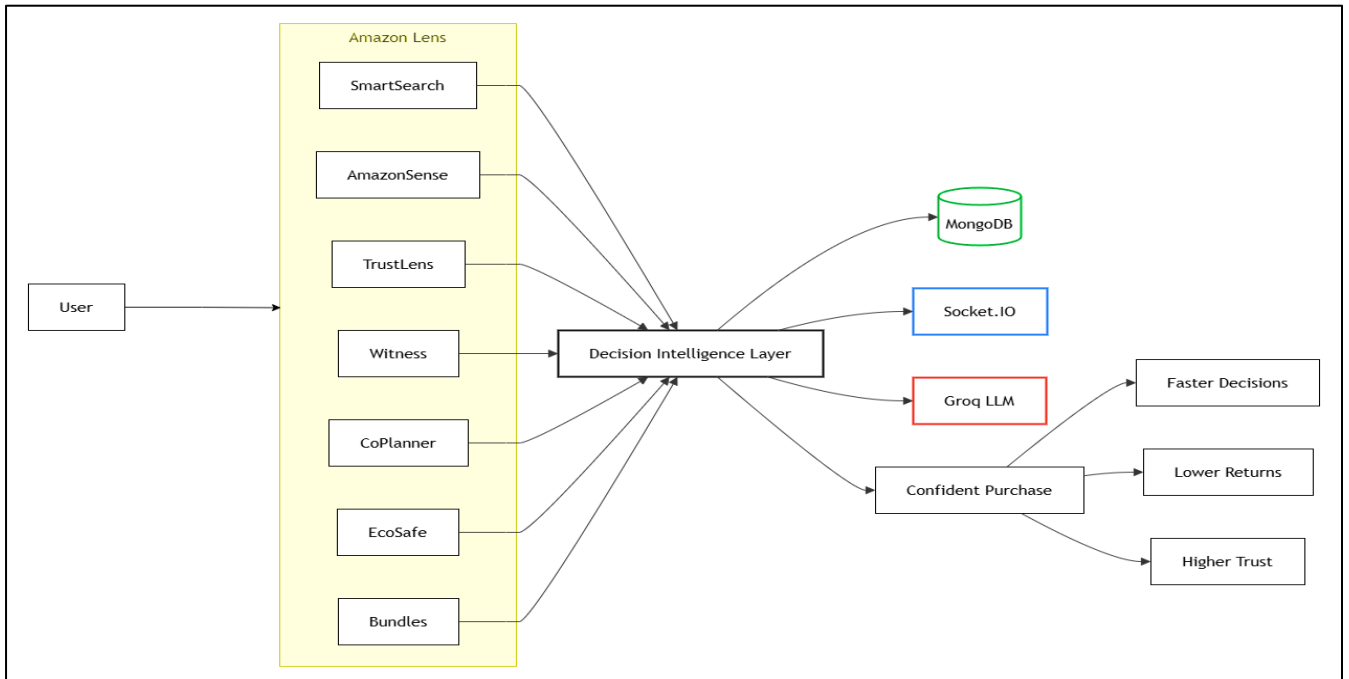
Amazon Lens Insight
Only 20% of your 50 scored purchases use sustainable packaging — this is the fastest area to improve.
Adding roughly 4 more eco-certified purchases could push your score closer to 85+.

Your Top Eco-Friendly Purchases

- EcoSmile Premium Bamboo Toothbrush (Pack of 4)
Personal Care - EcoSmile Retail **94**
Climate Leader
- Philips LED Study Lamp with Multi-Level Dimming
Lighting - Signify Innovations India **87**
Eco Advanced
- Wipro Garnet 9W LED Dimmable Bedside Lamp
Lighting - Wipro Consumer Lighting **82**
Eco Advanced
- Fire TV Stick 4K Max (2023) | Wi-Fi 6E | Alexa Voice Remote
Streaming Devices - Amazon **80**
Eco Advanced

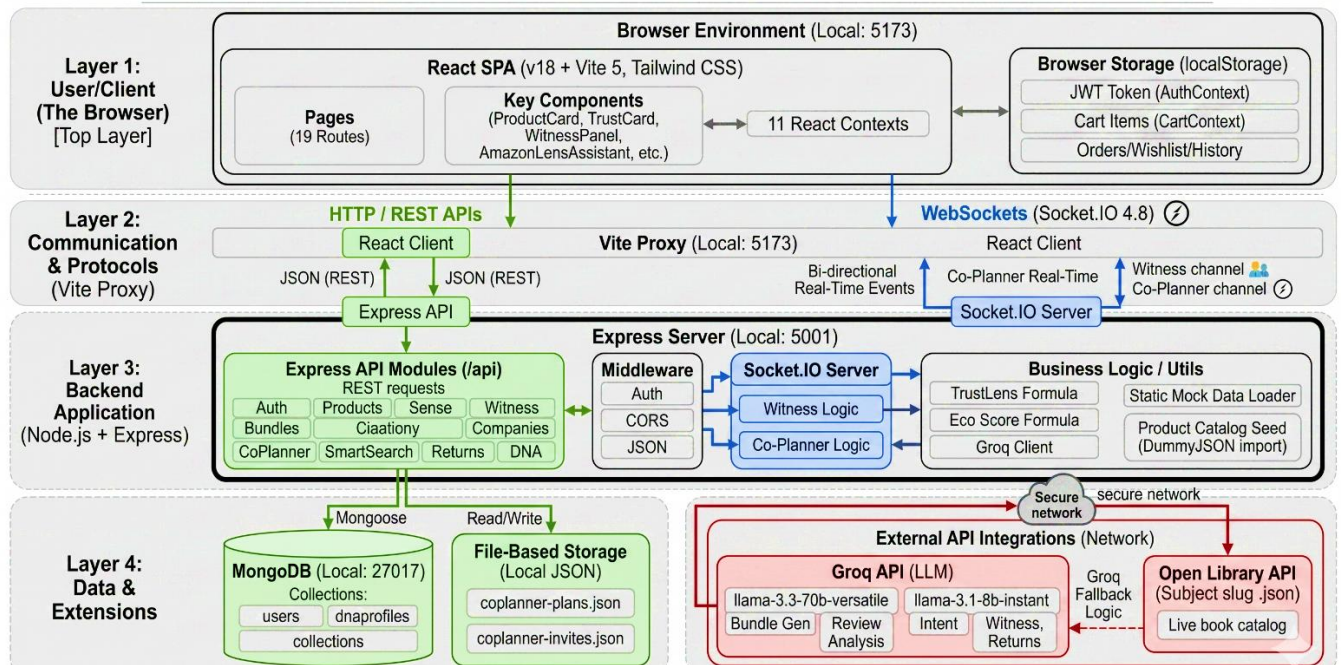
3. Tech Architecture & Scaling

Architecture



More Detailed:

AMAZON LENS — SYSTEM ARCHITECTURE DIAGRAM



Tech Stack

Layer	Technology	Why
Frontend	React 18, Vite 5, Tailwind CSS 3.3, React Router v6	Fast, responsive SPA with optimized build times.
Backend	Node.js, Express 4.18	Lightweight API layer supporting high concurrency
Database	MongoDB 7 (Mongoose)	Flexible NoSQL document store for dynamic data.
AI / LLM	Groq API — llama-3.3-70b-versatile , llama-3.1-8b-instant	Ultra-low latency inference for real-time agentic workflows, complex reasoning, and cost-effective, high-throughput text processing.
Auth	JWT (bcryptjs, 7-day, localStorage)	Stateless user authentication.
External APIs	Groq, Open Library, DummyJSON	Data sourcing for AI, books, and mock products.
Tooling	Vite, nodemon, concurrently	Unified client/server development workflow.

Key Algorithms & Complexity

1. Multi-Criteria Trust Consolidation (TrustLens Engine)

- **Algo:** Weighted linear model normalizes and aggregates 5 core metrics: verified ratings, return rates, reorders, seller track record, and delivery consistency.

$$\text{Equation: } T = \sum (w_i \times s_i)$$

- **Complexity:** $O(n)$, where n is the number of trust signals.
- **Rationale:** Transparent, explainable, and executes instantly without introducing database or API latency.

2. Behavioral Personalization (Amazon Sense)

- **Algo:** A tracking array applies asymmetric, decaying weights to live user actions:

$$\text{Purchase (1.0)} > \text{Wishlist (0.5)} > \text{Cart Add (0.4)} > \text{View (0.1)}$$

to dynamically update affinity profiles.

- **Complexity:** $O(k)$, where k is the stream of real-time user events.
- **Rationale:** Eliminates expensive ML model retraining, enabling instant, in-session personalization for urgent buyers.

3. Intent Understanding (Smart Search)

- **Algo:** A parsing pipeline extracts unstructured queries (e.g., "Study setup under ₹25k") into programmatic tokens: Intent, Budget, and Bundle options. Results are ranked via Amazon Sense vectors.
- **Complexity:** $O(m)$, where m is the filtered product candidate set.

- *Rationale:* Compresses manual, multi-step keyword browsing into a single-click, personalized checkout bundle.

4. Real-Time Matching (Witness)

- *Algo:* Uses a memory-mapped Redis hash map to track live socket connections.

The engine performs a direct key lookup for online, verified owners of a Product_ID and bridges them into a Socket.IO room.

- *Complexity:* **O(1)** constant time for room allocation and matching.
- *Rationale:* Guarantees sub-second chat connection speeds and prevents database bottlenecks during high global concurrent traffic.

Scaling Strategy

Amazon Lens follows a modular architecture where Smart Search, TrustLens, Amazon Sense, Witness, and Co-Planner can scale independently. Additional instances can be deployed behind load balancers without modifying application logic.

- **Stateless Scaling & Sharding:**

The architecture decouples and containerizes core servers into auto-scaling clusters. By utilizing MongoDB horizontal sharding with a hashed Product_ID key, write workloads are evenly split across regional nodes, eliminating data bottlenecks and supporting 1000x user growth.

- **Distributed Real-Time Infrastructure:**

To protect core APIs, all live connection layers are isolated. Shared *CoPlanner* carts and live *Witness* chats route through a Redis Pub/Sub adapter cluster, enabling sub-millisecond data synchronization across global nodes for millions of concurrent shoppers.

- **Edge-Cached Personalization:**

Calculating intensive *TrustLens* scores on every page load would freeze database operations. Instead, metrics are pre-computed and saved as static key-value hashes in geographically distributed Redis edge caches, dropping lookup speeds to a flat O(1) and slashing core database load by 90%.

4. Future Vision

Where This Goes

In 1–3 years, this platform evolves from a set of quick-commerce features into an *autonomous, zero-search shopping intelligence layer* that entirely *eliminates manual decision-making*. Instead of users digging through thousands of products and reading hundreds of contradictory reviews, the platform acts as a *predictive engine*. It *understands immediate intent*, *proactively manages group requirements*, builds instant trust through *verified community networks*, and *automates checkouts* down to a single click.

Roadmap

Horizon	Milestone	Impact
0-3 mo	Core Optimization & Precision Vetting: Integrate long-term product durability matrices into <i>TrustLens</i> and refine <i>Amazon Sense</i> using session-based behavioral drift tracking.	Projecting 1M+ active quick-commerce users ; targeting a 15% lift in checkout conversion rates and an immediate 20% reduction in cart abandonment metrics .
3-6 mo	Network & Collaborative Expansion: Scale <i>CoPlanner</i> to support persistent corporate, small business, and household profiles. Transition <i>Witness</i> into an incentivized, verified expert network.	Scale to 5M+ users ; drive an estimated 25% increase in Average Order Value (AOV) via natural language, multi-item programmatic bundling.
6-12 mo	Predictive & Anticipatory Commerce: Deploy <i>context-aware predictive algorithms</i> that surface dynamically compiled, individualized setups on the homepage <i>prior</i> to explicit user search entry.	Scale to 20M+ global users ; compress the end-to-end purchase pipeline to under 15 seconds , scaling system capacity to handle 10M+ automated bundles .

Multi-Segment Expansion

The underlying system architecture is designed as a modular intelligence engine, allowing seamless adaptation across diverse, high-value verticals:

- Phase 1: B2B Enterprise Procurement & Operations (Immediate Adjacent):**
 Transition *CoPlanner* and *Smart Search* into corporate supply chains. Teams can articulate complex operational requirements using natural language, allowing decentralized departments to collaborate on single, high-speed corporate carts governed by automated, internal budgetary compliance policies.
- Phase 2: Regulated Healthcare & Rapid Pharmacy (Mid-Term):**
 Adapt *TrustLens* and *Amazon Sense* to handle pharmaceutical options, medical equipment reliability metrics, and urgent prescription fulfilment. Users facing critical health scenarios

can utilize conversational inputs to instantly match complex needs with verified, regulatory-compliant medical solutions under strict delivery deadlines.

- **Phase 3: Hyper-Local On-Demand Services & Logistics (Long-Term):**

Export the real-time matching topology of *Witness* and *Amazon Sense* to the service economy. This maps user intent directly to localized, verified contractors, logistics providers, and on-demand technical experts based on immediate proximity and immutable performance scores.

Value Impact

- **Customer Decision Time:**

Amazon Lens reduces product validation time from an estimated **8–12 minutes of manual research** (reviews, seller checks, comparisons) to **under 60 seconds** through TrustLens, Witness, and Smart Search.

- **Return Reduction:**

By combining behavioural personalization (Amazon Sense) with trust scoring (TrustLens), the platform targets a **20–30% reduction in avoidable returns**, lowering reverse-logistics and support costs.

- **Conversion Growth:**

Intelligent bundles, personalized ranking, and natural-language intent understanding can drive an estimated **15–20% increase in conversion rates** and **10–15% higher average order value**.

- **Operational Efficiency:**

Trust scores, preference profiles, and search results are precomputed and cacheable, enabling **sub-second recommendation latency** while supporting millions of concurrent users.

- **Scalable Impact:**

As a marketplace intelligence layer, Amazon Lens can be deployed across **100M+ active shoppers**, improving trust, personalization, and purchase confidence without requiring changes to the underlying commerce infrastructure.

Links:

GitHub [URL] : <https://github.com/nityavarma123/AmazonLens>

Video: Available with submission